

BLUE HILL OBSERVATORY, MA, USA

WMO: 744907

Lat: 42.212N

Lon: 71.114W

Elev: 194

StdP: 99.02

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 14753

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-12.7	-23.8	0.4	-13.8	-21.4	0.6	-10.8	15.6	5.9	14.2	2.0	6.9	300	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	31.5	22.6	29.8	21.7	28.1	20.7	24.1	28.8	23.3	27.4	22.4	26.0	5.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.8	26.0	22.2	17.2	25.3	21.3	16.4	24.4	73.4	29.0	70.0	27.4	66.9	25.7	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.3	10.1	DB	-18.7	34.1	3.3	1.4	-21.1	35.1	-23.0	35.9	-24.9	36.6	-27.2	37.6	(4)
(5)			WB	-19.5	25.7	3.0	0.8	-21.6	26.2	-23.3	26.7	-25.0	27.1	-27.1	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-2.8	-1.5	2.3	8.7	14.2	18.9	22.4	21.8	17.9	11.6	6.1	0.6	(6)	
	DBStd	9.84	6.08	4.97	5.08	4.72	4.42	4.03	3.12	2.93	3.63	4.38	4.66	5.36	(7)	
	HDD10.0	1516	399	323	244	80	11	1	0	0	1	33	132	293	(8)	
	HDD18.3	3375	656	556	498	292	144	42	4	6	51	213	366	548	(9)	
	CDD10.0	1543	1	0	5	40	139	267	385	366	238	82	17	3	(10)	
	CDD18.3	362	0	0	0	2	15	58	132	112	38	4	0	0	(11)	
	CDH23.3	2479	0	0	1	32	149	429	960	697	199	13	0	0	(12)	
	CDH26.7	718	0	0	0	9	43	123	312	185	45	1	0	0	(13)	
(14) Wind	WSAvg	5.4	6.1	6.1	6.1	5.7	5.0	4.8	4.7	4.5	4.9	5.6	5.8	6.1	(14)	
(15) Precipitation	PrecAvg	1245	111	105	118	103	95	91	85	98	106	107	115	126	(15)	
	PrecMax	1762	295	237	332	263	232	440	271	213	314	374	248	320	(16)	
	PrecMin	651	25	18	8	30	14	4	17	14	18	10	13	29	(17)	
	PrecStd	223	60	46	64	53	51	83	48	53	66	71	59	66	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	13.7	20.3	27.9	31.0	32.4	33.8	32.6	30.8	25.1	19.2	17.0	(19)	
		MCWB	12.9	9.8	11.5	16.2	19.1	23.1	23.7	24.0	22.7	18.3	15.0	14.5	(20)	
	2%	DB	11.4	10.8	15.6	22.3	27.4	29.9	31.9	30.3	27.7	22.4	17.2	13.4	(21)	
		MCWB	9.5	8.3	9.5	12.9	17.8	21.9	23.0	22.3	20.4	17.4	13.9	11.5	(22)	
	5%	DB	7.9	7.7	12.2	18.8	24.1	27.7	30.0	28.7	25.7	20.2	15.0	11.1	(23)	
		MCWB	5.9	4.6	7.8	11.4	16.3	20.1	22.2	21.4	19.8	16.5	12.3	9.4	(24)	
	10%	DB	5.2	5.5	9.5	16.3	21.9	25.9	28.1	27.3	23.6	17.9	12.9	8.4	(25)	
		MCWB	3.0	2.6	6.2	10.4	15.5	19.2	21.2	20.6	18.9	14.6	10.2	6.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	11.7	13.9	17.5	21.2	24.6	25.7	25.1	23.5	20.8	16.9	15.2	(27)	
		MCDB	14.2	12.1	17.0	23.6	27.5	30.0	31.6	30.7	28.0	22.3	18.2	16.4	(28)	
	2%	WB	10.0	8.4	11.2	15.5	19.6	23.0	24.3	24.0	22.2	18.8	14.9	12.1	(29)	
		MCDB	11.1	10.4	14.1	20.6	25.5	28.4	29.2	28.0	25.2	21.1	16.2	13.4	(30)	
	5%	WB	6.5	5.2	8.7	13.4	18.0	21.8	23.3	23.1	21.3	17.1	12.9	9.7	(31)	
		MCDB	7.8	7.3	11.0	17.1	22.1	25.6	27.8	26.2	23.8	19.1	14.5	10.7	(32)	
	10%	WB	3.3	2.9	6.2	11.3	16.4	20.6	22.4	22.3	20.3	15.4	11.0	6.9	(33)	
		MCDB	4.9	4.7	9.2	14.7	20.1	24.1	26.4	25.2	22.8	17.9	12.6	8.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.2	8.7	9.8	9.9	9.5	9.4	9.0	8.8	8.2	7.7	7.2	(35)	
		MCDBR	10.3	10.4	12.3	14.1	13.8	12.0	11.2	10.5	10.6	10.3	9.4	9.8	(36)	
	5% WB	MCWBR	9.5	8.6	8.2	7.5	7.0	6.1	4.7	4.5	5.4	6.3	7.5	9.2	(37)	
		MCWBR	10.3	10.1	11.4	12.7	12.7	11.1	9.9	9.0	9.2	9.2	9.0	10.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.285	0.294	0.318	0.359	0.394	0.437	0.457	0.420	0.363	0.326	0.302	0.288	(40)	
		taud	2.512	2.502	2.496	2.407	2.344	2.252	2.204	2.293	2.434	2.554	2.581	2.576	(41)	
	Ebn at Noon		873	920	933	910	881	840	818	839	869	871	848	842	(42)	
		Edh at Noon	71	84	95	112	123	135	141	125	101	79	67	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.79	2.66	3.66	4.70	5.22	5.66	5.93	5.33	4.30	2.86	1.95	1.48	(44)	
	RadStd		0.12	0.23	0.26	0.50	0.57	0.54	0.35	0.25	0.30	0.25	0.15	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (72 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92	+41

Nomenclature: See separate page